MEDET JUMADILDAYEV

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EDUCATION

Nazarbayev University, Kazakhstan *Aug 2023 - June 2027 (Expected)*

B.S. in Mathematics

EXPERIENCE

Institute of Mathematics and Mathematical Modeling **Almaty, Kazakhstan**

Research Specialist *August 2025 - Present*

- Developed a generalization of the matching polynomial for edge-disjoint matchings, obtaining exact Gaussian integral representations and asymptotic enumeration results for complete multipartite graphs.
- Designed an efficient parallelized algorithm for computing N-commutators for Weyl and Poisson algebras.
- Implemented an algorithm that generates transfer matrices for combinations of monomino, domino, and L-tromino tilings to obtain generating functions and growth rates.

Institut de recherche mathématique avancée **Strasbourg, France**

Research intern under the supervision of Professor Vladimir Dotsenko *May 2026 - July 2026*

- Implemented parallelization and the triangle lemma for the Gröbner bases for operads Haskell software.
- Found a finite Gröbner basis for the $\mathcal{X}^-$ operad and a computational method to prove the structure of the infinite Gröbner basis for the $\mathcal{X}^+$ operad, where $\mathcal{X}^\pm$ are operads completing Loday's operadic butterfly diagram.

Institute of Smart Systems and Artificial Intelligence **Astana, Kazakhstan**

Research assistant under the supervision of Madina Abdrakhmanova *March 2024 - December 2024*

- Implemented training metrics for the vision transformer and multi-modal fusion approaches for Few-Shot learning.

PUBLICATIONS

[1] Medet Jumadildayev. "Increasing Trees and the Degree-Chromatic Polynomial". In: *Kazakh Mathematical Journal* 25.4 (Dec. 2025), pp. 57–66. DOI: 10.70474/ev8j4t48.

[2] Madina Abdrakhmanova, Assel Yermekova, Yuliya Barko, Vladislav Ryspayev, Medet Jumadildayev, and Huseyin Atakan Varol. "One Model to Rule Them all: A Universal Transformer for Biometric Matching". In: *IEEE Access* 12 (2024), pp. 96729–96739. DOI: 10.1109/ACCESS.2024.3426602.

[3] Askar Dzhumadil'daev and Medet Jumadildayev. "Euler tangent numbers modulo 720 and Genocchi numbers modulo 45". In: *Proceedings of the Japan Academy, Series A, Mathematical Sciences* 98.8 (2022), pp. 63–66. DOI: 10.3792/pjaa.98.012.

PREPRINTS

[1] Vladimir Dotsenko and Medet Jumadildayev. "Binomial coefficients and the operadic butterfly". Submitted to *The Journal of Combinatorial Algebra*. 2026. DOI: 10.48550/arXiv.2609.33480.

[2] Medet Jumadildayev. "Duality Relations of Graph Polynomials". Submitted to *Discrete Mathematics*. 2025. DOI: 10.48550/arXiv.2512.15351.

[3] Medet Jumadildayev. "Enumeration of multipartite series-reduced trees". Preprint. 2025. DOI: 10.48550/arXiv.2512.17216.

ACHIEVEMENTS

- Presented "Multipartite Labeled Series-Reduced Trees" at the 15th **ISAAC Congress**, 2025, Nazarbayev University, Astana.
- Third-degree diploma at **ICPC** 2023-2024 Northern Eurasia finals.
- Recipient of the **Yessenov Stipend** 2025 and the **Yessenov Science Internship Grant** 2026.
- Finalist at the **Regeneron International Science and Engineering Fair (ISEF)** 2021.
- First-degree diploma at the **Lomonosov 2026 Conference**, Moscow State University Astana branch.
- Third-degree diploma at the Kazakhstan National Olympiad 2022 in **Competitive Programming**.

SKILLS

Programming Languages: Python, C++, Haskell, Wolfram Mathematica
Technologies/Frameworks: OpenMP, SageMath, Pandas, NumPy, TensorFlow, Keras, PyTorch, Linux, Git
Languages: Kazakh (Native), Russian (Native), English (proficient), German (pre-intermediate), French (beginner).